

**CloudUSN
V100R020C60**

WSFD-121403 5G NSA User Traffic Charging

Applicable NEs

MME

Definition

The 5G NSA User Traffic Charging feature differentiates traffic for more accurate statistics and charging. Specifically, it enables the MME to identify and differentiate the user-plane IP addresses of the eNodeB and gNodeB therefore notifying the S-GW of the UE type (4G or NSA UE) belonging to incoming traffic. Additionally, the MME sends Notify Request messages to notify the HSS of UE type changes.

Benefits

For...	Benefits
Carriers	Identify and charge NSA traffic accurately in convergent 4G/5G networks.
Subscribers	Enjoy high-speed broadband wireless services and fair charging.

Application Scenarios

This feature is used for charging during evolution of carrier networks from 4G to 5G.

Carriers can upgrade EPC devices and deploy gNodeBs on the (R)AN in the early phase.

Availability

NEs Involved

NE	Supporting Version	Function
MME	V100R020C60	- Includes a private extension IE in an MBR message to indicate an NSA bearer towards the S-GW. - Identifies subscriber attribute changes, notifying the HSS through the RAT type in the Notify Request message.
S-GW/P-GW	All	Identifies the UE access type based on traffic usage.
UE	All	Supports NSA access.

License Requirement

This feature is optional and requires a license. The license control item is 82200DCF LKV2RAFDIFF01 5G NSA User Traffic Charging.

Interaction with Other Features

Interaction Type	Feature	License Control Item	Description
Dependencies	WSFD-021100 Supporting 5G NSA(Opt.3) Networking	-	This feature takes effect only when WSFD-021100 Supporting 5G NSA(Opt.3) Networking is activated.
Dependencies	WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management	-	This feature takes effect only when WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management is activated.

Impact on the System

None

Application Limitations

- This function applies only to Option 3a/3x networking. It does not apply to Option 3 networking, where the core network is unaware of any UE traffic steered to the gNodeB and traffic is steered by the eNodeB. Traffic distribution must be disabled on the (R)AN side to prevent statistical errors.
- The initial message sent by the eNodeB must contain its user-plane IP address.
- This function cannot take effect when the eNodeB and gNodeB use the same user-plane IP address.

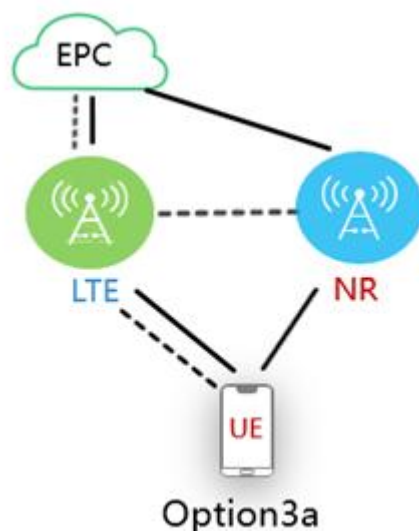
- Huawei devices must be deployed as the MME, S-GW, P-GW, and HSS, as a customized extension IE is required.

Implementation Principles

Networking Structure

Huawei uses Option 3/3a/3x as the networking solution for early deployments of 5G NSA. In these options, the control-plane signaling is transmitted between the UE and the MME through the eNodeB. User-plane data flows are steered to both eNodeBs and gNodeBs and finally to UEs. User-plane traffic steering varies with Option 3/3a/3x. This feature applies only to Option 3a/3x.

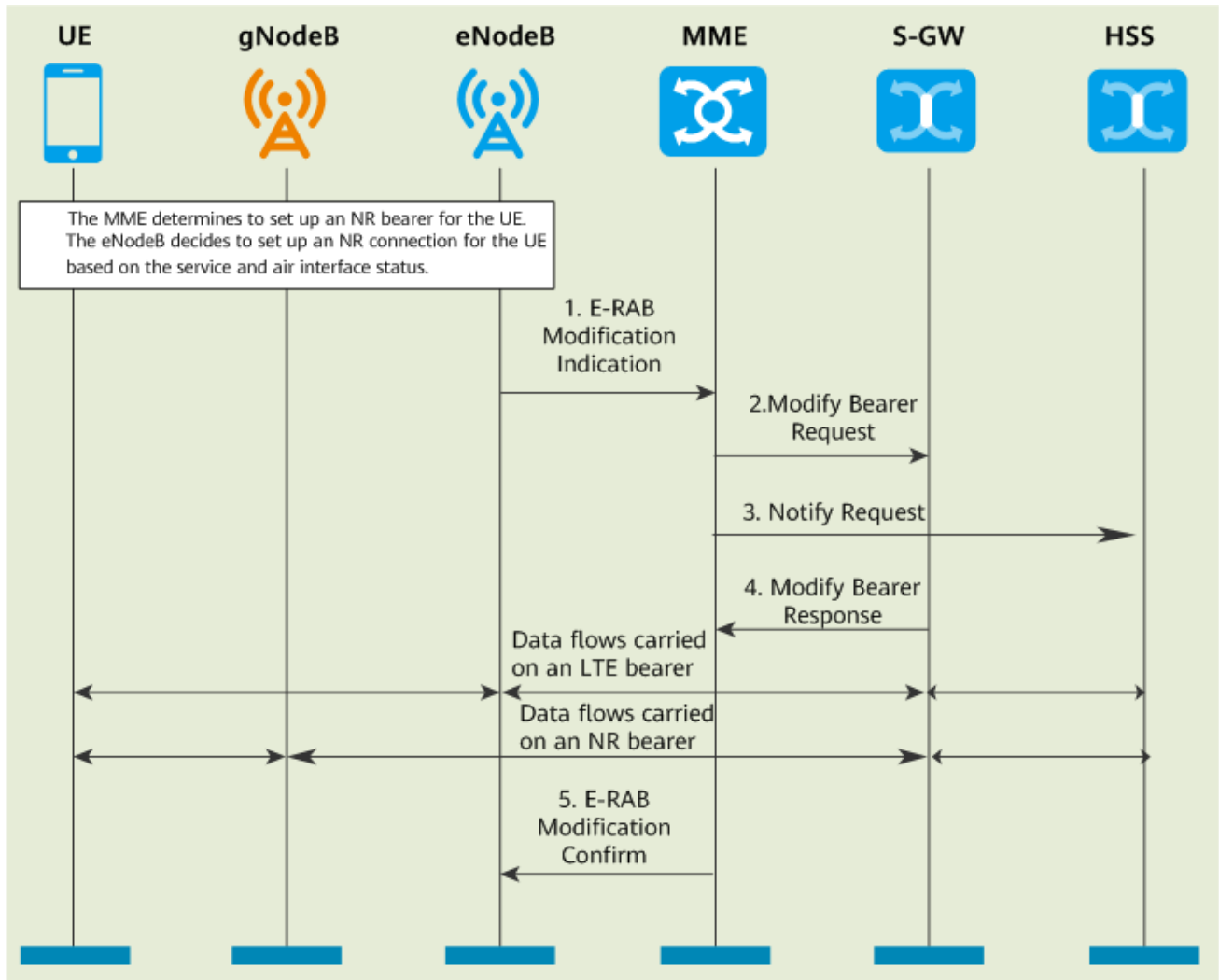
- In Option 3a, an eNodeB initiates an E-RAB Modification Indication procedure to establish an S1-U transmission path between the gNodeB and the S-GW. The eNodeB then switches the bearer requiring data steering to the established S1-U path, and the S-GW steers data to the eNodeB and gNodeB.
- In Option 3x, an S1-U path is established between the gNodeB and the S-GW, similar to Option 3a. But unlike 3a, the gNodeB can directly establish a user-plane transmission path to the eNodeB to steer user-plane data.



Implementation Principles

- After the traffic differentiation function is enabled on the MME by running **SET NSACTRL**, the MME learns the user-plane IP address of the eNodeB carried in the Initial Context Setup Response message in the Attach, Tracking Area Update (TAU), or Service Request procedures. Such procedures enable idle UEs to enter the connected state.
- The eNodeB sends an E-RAB Modification Indication message to the MME. The message contains the IP address of the gNodeB to which bearer data will be steered. If the MME finds that the user-plane IP address received in the message belongs to the gNodeB, not the learned user-plane IP address of the eNodeB, it notifies the S-GW by sending a Modify Bearer Request message.

Figure 1-1 E-RAB modification indication procedure



1. The eNodeB determines to establish an NR bearer for a UE based on the current service type of the UE, air interface status, and other information. The eNodeB then sends the MME an E-RAB Modification Indication message with the IP address of the to-be-split bearer set to the gNodeB IP address.
2. The MME sends a Modify Bearer Request message to the S-GW, instructing the S-GW to switch the user-plane transmission path from the eNodeB to the gNodeB or vice versa. The message carries the IP address of the target base station (eNodeB or gNodeB) and the extension IE indicating whether the IP address is the gNodeB IP address.
3. (Optional) The MME sends a Notify Request message to notify the HSS that the UE is an NSA/4G UE.
4. The S-GW updates the bearer IP address and sends the update result to the MME through a Modify Bearer Response message.
5. The MME notifies the eNodeB of the update result through an E-RAB Modification Confirm message.

Charging and CDRs

None

Standards Compliance

Category	Number	Name
3GPP	23.401 V15.6.0	General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network(E-UTRAN) access
	29.274 V15.3.0	Evolved GPRS Tunnelling Protocol for Control Plane (GTPv2-C)

Release History

Feature Version	Product Version	Details
01	V100R020C60	First release.

Related Topics

Feature Deployment

- 1.1 Activating the 5G NSA User Traffic Charging Feature
- 1.2 Deactivating the 5G NSA User Traffic Charging Feature

Commands

- Set NSA Control Parameters (SET NSACTRL)
- Add IMSI-based Diameter Compatibility Parameters (ADD DMCMPYBYIMSI)
- List IMSI-based Diameter Compatibility Parameters (LST DMCMPYBYIMSI)
- Modify IMSI-based Diameter Compatibility Parameters (MOD DMCMPYBYIMSI)
- Remove IMSI-based Diameter Compatibility Parameters (RMV DMCMPYBYIMSI)
- Set Compatibility of Diameter Interface (SET DMCMPY)
- List Compatibility of Diameter Interface (LST DMCMPY)

Alarms

- ALM-80631 Single Process Resource Congestion
- ALM-80632 Single Process Resource Overload

Performance Counters

If bit 7 of BYTE_EX_B41 is set to **1**, the values of the following counters may decrease:

- 117480544 Total Number of Attached NSA UEs in S1 Mode
- 117480550 Total Number of Bearers Activated for NSA UEs in S1 Mode
- 117480547 Total Number of PDN Connections Activated for NSA UEs in S1 Mode

Software Parameters

- BIT3 of BYTE_EX_B41 Controlling Whether the MME Includes the Private Extension IE in a Modify Bearer Request Message Sent to the S-GW to Notify the S-GW Whether a Bearer Is an NSA Bearer
- BYTE_EX_B44 Specifying the Scope of S-GWs That Support the Modify Bearer Request Message Containing the Private Extension IE
- BIT7 of BYTE_EX_B41 Controlling How to Measure the Numbers of NSA UEs, NSA PDN Connections, and NSA Bearers

[13.1 Activating the 5G NSA User Traffic Charging Feature](#)

[13.2 Deactivating the 5G NSA User Traffic Charging Feature](#)

1.1 Activating the 5G NSA User Traffic Charging Feature

Scenarios

The 5G NSA User Traffic Charging feature differentiates traffic for more accurate statistics and charging. Specifically, it enables the MME to identify and differentiate the user-plane IP addresses of the eNodeB and gNodeB therefore notifying the S-GW of the UE type (4G or NSA UE) belonging to incoming traffic. Additionally, the MME sends Notify Request messages to notify the HSS of UE type changes.

This section describes how to active the 5G NSA User Traffic Charging feature.

NOTE

This feature is used only with the MME.

Impact on the System

This operation has no adverse impact on the system.

Prerequisites

Conditions

- You have logged in to the HUAWEI Operation & Maintenance System.
- WSFD-021100 Supporting 5G NSA(Opt.3) Networking and WSFD-021101 5G NSA (Opt.3) Dual Connectivity Management have been deployed.
- The license for this feature has been loaded.

Data

Table 1-1 Data required

Category	Parameter	Example Value	How to Obtain	Command
NSA control parameters	Traffic Differentiator (TRAFFICDIFF)	YES	Planned on the entire network	SET NSACTRL
	Method for MME Notify HSS of NSA UE (NRTYPE)	IPTYPE	Planned on the entire network	
Policy of reporting the RAT type in a Notify Request message	SUBRANGE	IMSI_PREFIX	Planned on the entire network	ADD DMCMPBYIMSI
	IMSIPRE	460123		
	NORRAT	YES		
Software parameters	Para type (DT)	BYTE_EX_B	Planned on the entire network	SET SOFTPARA
	Para index (PARANUM)	44	Planned on the entire network	
	Para value (VALUE)	0	Planned on the entire network	
Bit of the software parameter table	Para type (DT)	BYTE_EX_B	Planned on the entire network	SET SOFTPARAOFBIT
	Para index (PARANUM)	41	Planned on the entire network	
	Para value (VALUE)	1	Planned on the entire network	
	Bit position (POSITION)	7	Planned on the entire network	
Adding an IP address group	IP group ID (IPGPID)	1	Planned on the entire network	ADD IPGP
	IP Group Name (NAME)	DCNR_SGW	Planned on the entire network	

Category	Parameter	Example Value	How to Obtain	Command
Addition of an IPv4 group member	IP Group ID (IPGPID)	1	Planned on the entire network	ADD IPV4GPMEM
	IPv4 Address (IPV4)	172.22.5.50	Planned on the entire network	
	IPv4 Subnet Mask (IPV4MSK)	255.255.255.0	Planned on the entire network	
Addition of an IPv6 group member	IP Group ID (IPGPID)	1	Planned on the entire network	ADD IPV6GPMEM
	IPv6 Address (IPV6)	2001:db8::	Planned on the entire network	
	IPv6 Address Prefix Length (IPV6MSK)	64	Planned on the entire network Planned on the entire network	

Procedure

Step 1 Configure the policy of carrying 5G UE information in the Modify Bearer Request message.

1. Configure whether the MME sends a private IE to the S-GW.

In the **MML Command - OMO_VNFC** window, run the following command:

SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=1, POSITION=3;

NOTE

0: The private IE is not carried.

1: BYTE_EX_B 44 (0-255) controls whether the private IE is carried.

2. Configure the S-GWs to which the MME sends private IEs.

SET SOFTPARA: DT=BYTE_EX_B, PARANUM=44, VALUE=0;

NOTE

0: The MME includes a private IE in a message sent to all S-GWs.

1-255: The S-GW includes a private IE in an IPGPID IE.

3. Configure an IP group for the S-GW.

Add an IP address group.

In the **MML Command - OMO_VNFC** window, run the following command:

ADD IPGP: IPGPID=IP Group ID, NAME=IP Group Name;

- Add an IPv4 group member.

In the **MML Command - OMO_VNFC** window, run the following command:

ADD IPV4GPMEM: IPGPID=IP Group ID, IPV4=IPv4 Address, IPV4MSK=IPv4 Subnet Mask;

- Add an IPv6 group member.

In the **MML Command - OMO_VNFC** window, run the following command:

ADD IPV6GPMEM: IPGPID=IP Group ID, IPV6=IPv6 Address, IPV6MSK=IPv6 Subnet Mask;

Step 2 (Optional) Configure Diameter compatibility. To configure the MME to include the RAT Type IE in the Notify Request message for home subscribers, foreign subscribers, and subscribers with a specified IMSI prefix, perform the following steps:

1. View the Diameter compatibility configurations.

 NOTE

Parameters other than **Subscriber Range**, **Network Operator Identity**, and **IMSI Prefix** in **ADD DMCMPBYIMSI** and **SET DMCMP** have the same functions. If both commands are configured, the system preferentially matches the configurations in **ADD DMCMPBYIMSI**. As such, before running **ADD DMCMPBYIMSI**, run **LST DMCMP** to list the configuration records of **SET DMCMP** to ensure that the default parameter settings of **ADD DMCMPBYIMSI** do not affect the existing system configurations.

In the **MML Command - OMO_VNFC** window, run the following command:

LST DMCMP: ;

2. Enable the MME to include the RAT Type IE in the Notify Request message sent to the HSS.
 - The software parameter BYTE_EX_B35 has been added to perform flow control on the Notify Request messages for NSA UEs sent to the HSS.

SET SOFTPARA: DT=BYTE_EX_B, PARANUM=35, VALUE=Para value;

- Configure the MME to include the RAT Type IE in the Notify Request message sent to the HSS.

ADD DMCMPBYIMSI: SUBRANGE=Subscriber Range, IMSIPRE=IMSI Prefix, NORRAT=SUPPORT;

 NOTE

In actual deployment, determine the configuration based on whether the HSS supports the RAT Type IE carried in the Notify Request message and whether the HSS needs to identify 5G NSA UEs. The following commands are used as examples:

- To enable all HSSs to support this feature, run the following command:

SET DMCMP NORRAT=SUPPORT;

- When all HSSs on the home network support this feature and none of the HSSs on the external networks support this feature, run the following command:

ADD DMCMPBYIMSI: SUBRANGE=HOME_USER, NOID=0, NORRAT=SUPPORT;

- To enable only the HSSs serving the UEs with the IMSI prefix 46123 or 46321 to support this feature, run the following commands:

ADD DMCMPBYIMSI: SUBRANGE=IMSI_PREFIX, IMSIPRE="46123", NORRAT=SUPPORT;

ADD DMCMPBYIMSI: SUBRANGE=IMSI_PREFIX, IMSIPRE="46321", NORRAT=SUPPORT;

Step 3 (Optional) Configure the MME to include the RAT Type IE in the Notify Request message sent to the HSS.

In the **MML Command - OMO_VNFC** window, run the following command:

SET NSACTRL: NRTYPE=IPTYPE;

Step 4 Configure the MME to support the 4G/5G subscriber traffic differentiation function.

In the **MML Command - OMO_VNFC** window, run the following command:

SET NSACTRL: TRAFFICDIFF=YES;

Step 5 Configure the policy for counting the number of NSA UEs and bearers on the MME.

In the **MML Command - OMO_VNFC** window, run the following command:

SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=VALUE_1, POSITION=7;

NOTE

Set this bit to **1** if the MME is required to measure the numbers of NSA UEs, PDN connections, and bearers based on the type of the gNodeB or eNodeB address used by the bearer of a UE when the (R)AN address type differentiation function is enabled.

This software parameter takes effect 24 hours after the function controlled by it is enabled. The MME learns the eNodeB address within 24 hours, preventing a large number of UEs from being mistakenly counted as NSA UEs. If the function controlled by the software parameter is disabled and then enabled, the setting takes effect after 24 hours.

----End

Verification

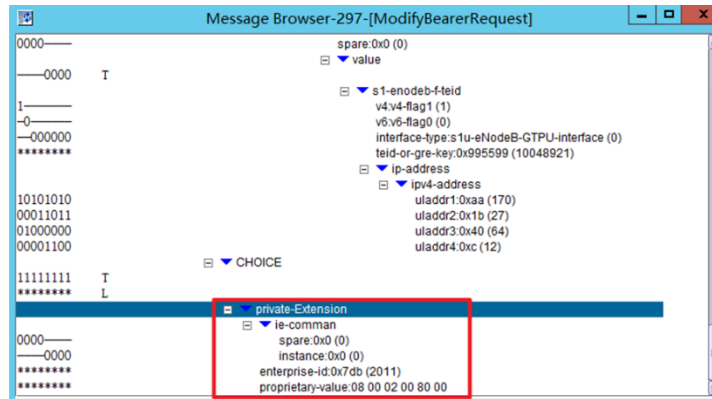
After this feature is configured on the related NEs, perform the following operations to check whether this feature is applied successfully:

Verify that the MME can match the IP address based on the procedure and notify the HSS and S-GW.

1. Create a user trace task on the MME and set **IMSI** to the IMSI of UE A to be traced.
2. Enable the UE to initiate a service to the 5G network to trigger the eNodeB to establish an NR bearer for the UE.
3. Enable UE A to attach to the MME and query for the user trace result.

Expected result:

- a. The E-RAB Modification Indication message contains the gNodeB IP address.
- b. The Modify Bearer Request message carries the gNodeB IP address and the extension IE indicating whether the IP address of the bearer is the gNodeB IP address. As shown in the following figure, bearer 5 uses the user-plane IP address of the gNodeB for communication.



- c. (Optional) The RAT Type IE in the Notify Request message indicates that the UE is an NSA UE.

Example

Task Description

Activate the 5G NSA User Traffic Charging feature.

Scripts

//Configure the policy of carrying 5G UE information in the Modify Bearer Request message.

```
//Configure whether the MME sends a private IE to the S-GW.
SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=1, POSITION=3;
//Specify the S-GWs to which private IEs are sent.
SET SOFTPARA: DT=BYTE_EX_B, PARANUM=44, VALUE=0;
//Add an IP group.
ADD IPGP: IPGPID=1,NAME="DCNR_SGW";
Add an IPv4 group member.
ADD IPV4GPMEM: IPGPID=1, IPV4="172.22.5.50", IPV4MSK="255.255.255.0";
Add an IPv6 group member
ADD IPV6GPMEM: IPGPID=1, IPV6="2001:db8::", IPV6MSK=64;
```

//(Optional) Configure the compatibility policy for the RAT Type IE in the Notify Request message for a UE with a specific IMSI prefix.

```
LST DMCMP;
ADD DMCMPBYIMSI: SUBRANGE=IMSI_PREFIX, IMSIPRE="46003", NORRAT=SUPPORT;
```

//(Optional) Enable the MME to include the RAT Type IE in the Notify Request message.

```
SET NSACTRL: NRATYPE=IPTYPE;
```

//Configure the MME to support 4G/5G UE traffic differentiation.

```
SET NSACTRL: TRAFFICDIFF = YES;
```

//Configure the policy for counting the number of NSA UEs and bearers on the MME.

```
SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=VALUE_1, POSITION=7;
```

1.2 Deactivating the 5G NSA User Traffic Charging Feature

Scenarios

This section describes how to deactivate the 5G NSA User Traffic Charging feature.

NOTE

This feature is used only with the MME.

Impact on the System

This feature has no adverse impact on the system.

Prerequisites

Conditions

- You have logged in to the HUAWEI Operation & Maintenance System.
- The 4G/5G traffic differentiation function has been activated.

Data

No additional data.

Procedure

Step 1 Disable the 4G/5G subscriber traffic differentiation function.

In the **MML Command - OMO_VNFC** window, run the following command:

SET NSACTRL: TRAFFICDIFF=NO;

Step 2 Configure the policy for counting the number of NSA UEs and bearers on the MME.

In the **MML Command - OMO_VNFC** window, run the following command:

SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=VALUE_0, POSITION=7;

Step 3 (Optional) Disable the MME from including the RAT Type IE in the Notify Request message.

SET NSACTRL: NRTYPE=OFF;

Step 4 Configure whether the MME sends a private IE to the S-GW.

SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=0, POSITION=3;

----End

Example

Task Description

Deactivate the 5G NSA User Traffic Charging feature.

Scripts

//Remove the NSA traffic differentiation configuration.

```
SET NSACTRL: TRAFFICDIFF=NO;
```

//(Optional) Configure the policy for counting the number of NSA UEs and bearers on the MME.

```
SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=VALUE_0, POSITION=7;
```

//(Optional) Disable the MME from including the RAT Type IE in the Notify Request message.

```
SET NSACTRL: NRTYPE=OFF;
```

//Configure whether the MME sends a private IE to the S-GW.

```
SET SOFTPARAOFBIT: DT=BYTE_EX_B, PARANUM=41, VALUE=0, POSITION=3;
```